

Myths

- Find more bugs
- Eliminate or reduce manual testers
- Unrealistic expectations
- Lack of planning
- Lack of a process

Solutions / Suggestions

- Avoid false set of expectations
- Selection of the right tool for the application
- Limitations of automation
- Separate the facts from the fictions
- Spend more time up front for research

When To Automate Tests

- Tests that will run with each new version
- Sanity tests
- Data-driven tests
- Test cases with higher complexity
- Higher manual execution time
- Regression

Types of Tests Automatable

Functional

- System
- Functional
- Sanity
- Regression
- UI Testing
- Usability
- Compatibility
- Cross Browser
- UAT
- E2E

Non-Functional

- Stress
- Load
- Performance

Coding

- UNIT
- Integration
- White Box

When Not To Automate Tests

- More manual involvement nature
- Tests that you only have to run once
- Tests that you need to run immediately
- Tests with no predictable results
- Lack of a process
- Strategic Development

Types of Tests Not Automatable

Functional

- Adhoc
- Exploratory
- User Experience
- Random testing

Coding

- Installation testing
- Setup testing
- Component Integration

Benefits of Automated testing

- Production of a reliable system
- Improvement of the quality of the test effort
- Reduction of the test effort
- Minimization of the schedule
- Testing more complicated and large applications

Automation Testing Lifecycle

1-5 preparation activity

6-8 Scripting

- Application & Business Study
- Feasibility Study
- Define scope of automation
- Identify the Automation tool
- Identify Framework
- Design and development of a test suite
- Continuous test execution
- Maintain test suite

Few Risks associated with Automation

- Skilled Workforce
- High initial cost of automation
- Constantly changing UI
- Unstable application

SELENIUM

About Selenium

- Selenium is a portable framework for testing web applications
- It is an open-source web-based automation testing suite
- It is not just a single tool but a suite of software's

Suite----Not a single tool but a group of packages for testing



Evolution

- Selenium Core
- Selenium Remote Control
- Selenium Grid
- Selenium IDE
- Selenium WebDriver
- Selenium 2

Selenium grid----run test cases parallelly in different machines, how we divide test cases

IDE----How to work with multiple browsers

Latest selenium version----4

Benefits of Selenium

- Flexibility
- Supports various operating systems, browsers and programming languages
- Supports parallel test execution that reduces time and increases the efficiency
- Selenium commands are categorized in terms of different classes

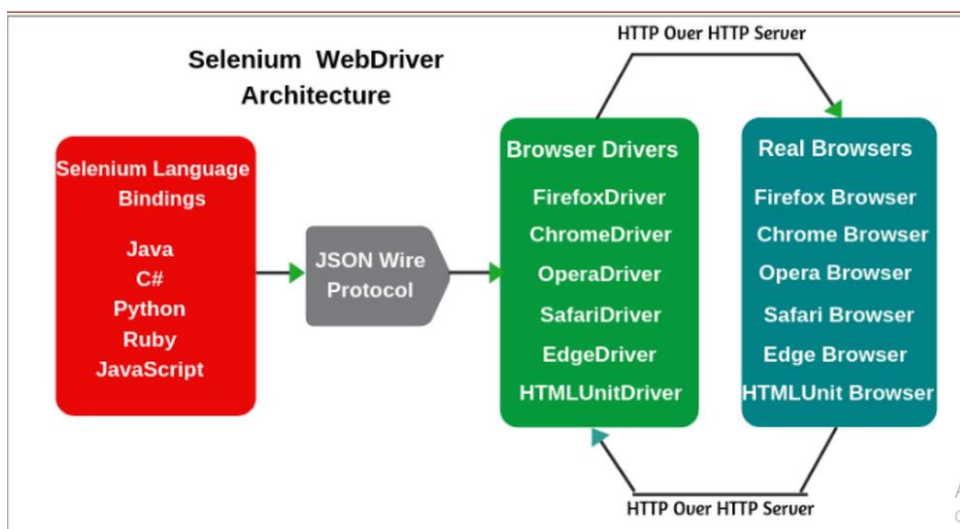
Selenium Limitations

- Selenium does not support desktop applications.
- Needed support from community forums to get the technical issues resolved
- Should know at least one of the supported programming languages
- Does not have any inbuilt reporting capability
- Not possible to perform testing on images

80% of test automation uses java → As selenium and java works very well and used for a very long time

Before versions did not have image testing so they added plugins into selenium to test images properly- Now Selenium is capable of image testing

Selenium Architecture:



Driver is separate jar file in previous versions now it is inbuilt in Selenium



WebDriver

- Selenium WebDriver is a browser automation framework that accepts commands and sends them to a browser
- It is implemented through a browser-specific driver
- It controls the browser by directly communicating with it
- It is a tool that automates the browser to mimic real user actions on the web
- Test scripts can be developed using any of the supported programming languages
- Can be run directly in most modern web browsers
- Languages supported by WebDriver include C#, Java, Perl, PHP, Python and Ruby
- Most of the commands used in Selenium WebDriver are simple & easy to implement (Commands are simply **methods** written in Java language)
- Handles Navigation, Alerts / Dropdowns, Switching between windows, etc

Activate Windows

Activate Windows

WEB ELEMENT



WebElement

Finding Element

- Element is a field, object, or any item in website
- Element in a web browser is identified using “Locators”
- Locators are
 - Attributes
 - Text
 - Tag Name
 - Dynamic Locator

Activate Windows

Locators

- Attribute - Each element has 3 important attributes:
 - Id
 - Name
 - Class
- Text
 - Link Text
 - Partial Link Text
- Tag Name
- Dynamic Locator – XPATH
 - Absolute XPATH
 - Relative XPATH

Why so many locators?

1. The element cannot always be located with id
2. Every element may not have id as locator any one of the locator may be available....so when we have all options we can choose an alter option when one is not working

What should a script contain?

- Ensure relevant packages / classes are imported
- Invoke a browser
- Create instance of the Web Driver
- Maximize the screen window
- Give the URL of the webpage & launch the website
- Wait till the web page loads
- Find each Web Element you want to access
- Invoke an action on each of those Web Element
- Finally close the Web Driver

Commands

WebDriver commands

- Close()
- Equals()
- findElement()
- get()
- getTitle()
- getWindowHandle()
- manage()
- navigate()
- quit()
- switchTo()
- wait()

WebElement commands

- Clear()
- Click()
- findElement()
- getAttribute()
- getClass()
- getLocation()
- getSize()
- getTagName()
- getText()
- isDisplayed()
- isEnabled()
- notify()
- sendKeys() Activate Windows
Go to Settings to activate Windows
- submit()
- wait()

SELENIUM-PART 2

XPATH

- A query language used for navigation thru XML documents to locate different elements
- A strategy to locate elements in Selenium
- Starts with double slash
- Uses single slash between each condition
- Uses double slash to by-pass parent tags
- For attribute use @
- While searching value, keep in mind that value is case & space sensitive
- Use and / or operators to give multiple conditions
- Multiple conditions may use different attribute or same attribute in conditions; and / or can also be used with attribute & text
- "Not" operator – used to find an attribute not existing in the tag
- Uses "contains" function
- x-path index starts with 1

XPATH

Syntax & example:

```
//TagName  
//TagName[index]  
//TagName[@attribute]  
//TagName[@attribute = 'AttributeValue']  
//TagName[text()]  
//TagName[text() = 'TextValue']
```

Using Functions :

```
//TagName[contains(text(),'Value')]  
//TagName[contains(@attribute,'Value')]  
//TagName[contains(string(),'Value')]
```

Using Operators:

```
//TagName[@attribute='Value1' and @attribute = 'Value2']  
//TagName[@attribute1='Value' or @attribute2 = 'Value']  
//TagName[not(@attribute)]  
//TagName[not(@attribute='Value')]
```


XPATH

Axes: use of relationship:

- Child, parent, following-sibling, preceding-sibling, ancestor, descendant
- Use scope operator "::" ; Wherever you give relation, give :: next to it
- While using condition, come from parent tag; means top down traverse
- Traverse can be done only to next nearest relationship

Syntax:

//TagName[relation::relativeTagName]

[xpath]/relation::TagName

Two Types of XPATH:

1. Absolute XPATH – traversing from one tag to next closest tag, top to bottom via the tree structure
Syntax: //TagName1/TagName2/TagName3
2. Relative XPATH – uses relationship
Syntax: //TagName[relation::relativeTagName]

WINDOW FORMATS

Web Page Windows

- External / Internal web page formats that carries messages
- Warnings or Business Information
- Mandates user to access & respond

Three Types

1. Alert (message box)
2. Popup (a window)
3. Ad (a banner)

Alert and popup hides the webpage but ads doesnot hide webpages so in selenium we have to handle alert and popups mostly

Alerts

- Small fixed size message box appears over web page
- Notified user with warning, information or error messages
- Prompts user to respond to proceed further
- Asks for permission to perform certain tasks

Three Types of Alerts:

1)Simple Alert

- ✓ Displays some information or warning on the screen.

2)Confirmation Alert.

- ✓ Asks permission to do some type of operation

3)Prompt Alert.

- ✓ Asks some input from the user and further action

Methods:

accept()
dismiss()
sendKeys()
getText()

PopUps

- A browser window that overlays a website on a screen, usually intentionally, and offers a marketing message to engage the visitor
- Can appear as a small window, a full-screen takeover, a window that slides in from the side, or an opt-in bar
- It appears on a website while a user is browsing
- Can be used to display marketing messages, notifications, surveys, navigations or errors
- User can interact with the pop-up by clicking on a button, link, or form.
- Pop-up doesn't disappear until the visitor closes it

How to Handle?

Measure the duration it takes to appear

Wait & locate the window

Proceed with Action

Wait is required to handle popups

Waits

- Most of the web applications are developed using Ajax and Javascript.
- When a page is loaded by the browser the elements may load at different time intervals

Selenium Web Driver Waits

1. Implicit Wait
2. Explicit Wait
3. Fluent Wait

Implicit Wait in Selenium

- Used to tell the web driver to wait for a certain amount of time before it throws a "NoSuchElementException".
- If the element is not located on the web page within that time frame, it will throw an exception

Explicit Wait in Selenium

- Used to tell the Web Driver to wait for certain conditions (Expected Conditions) or maximum time exceeded before throwing "ElementNotVisibleException" exception
- Intelligent kind of wait, but it can be applied only for specified elements.
- Gives better options than implicit wait as it waits for dynamically loaded Ajax elements

Expected Conditions:

1. alertIsPresent()
2. elementSelectionStateToBe()
3. elementToBeClickable()
4. elementToBeSelected()
5. frameToBeAvaliableAndSwitchToIt()
6. invisibilityOfTheElementLocated()
7. invisibilityOfElementWithText()
8. presenceOfAllElementsLocatedBy()
9. presenceOfElementLocated()
10. textToBePresentInElement()
11. textToBePresentInElementLocated()
12. textToBePresentInElementValue()
13. titles()
14. titleContains()
15. visibilityOf()
16. visibilityOfAllElements()
17. visibilityOfAllElementsLocatedBy()
18. visibilityOfElementLocated()

AJAX / AutoComplete

- ✓ AJAX stands for Asynchronous JavaScript & XML
- ✓ A technique used for creating fast and dynamic web pages
- ✓ It is asynchronous and uses a combination of Javascript and XML
- ✓ Allows the Web page to retrieve small amounts of data from the server or updates the part/s of a web page without reloading the entire page
- ✓ Sometimes it may load in a second and sometimes it may take longer. We have no control over loading time.
- ✓ AJAX sends HTTP requests from the client to server and then process the server's response without reloading the entire page
- ✓ We wouldn't know when the AJAX call would get completed and the page has been updated
- ✓ Handling Auto Suggestion or Auto Complete Textbox in Web Based Applications
- ✓ Autocomplete or Auto Suggestion allows the browser to predict the value
- ✓ When a user starts to type in a field, the browser should display options to fill in the field, based on earlier typed values.
- ✓ The auto complete works with the following input types: text, search, url, email and datepickers

21/04/2025

SELENIUM-Page Object model:

- For each page where we need automation control, we create an object repository with all fields in the page as an object
- For each page we can create different script-test cases which uses different object repository (reusable component)